Arnav Ab Maiti

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.14
The New Tulip International School	Class XII (CBSE)	2023	95.00%
The New Tulip International School	Class X (CBSE)	2021	95.80%

SCHOLASTIC ACHIEVEMENTS

- KVPY Scholar: **Secured All India Rank 495** in KVPY
- Olympiads - Qualified for **INMO 2022** from Gujarat, Top 1% performers in **NSEC 2022**
- Secured rank **772** in IICPC Quantfest 2026 out of 10000+ participants
- Finished **87th** in **Optiver Trade-A-Thon** among 400+ participants across different IIT's
- Received **Academic Excellence Scholarship** from SOF for consistently being among the top performers nationally
- LogIQids: **Winner in State Level** in LogIQids Logical Reasoning Olympiad 2021

WORK EXPERIENCE

Quant Research Intern | 26 Miles Capital

- Researched intraday mean-reversion signals on NIFTY 50 futures OHLCV data -- built and tested multiple approaches including **VWAP z-score, ADX regime detection, RSI divergence, and logistic regression** with walk-forward IC analysis to identify high-probability reversal zones.
- Built an **options analytics pipeline** computing Greeks (delta, gamma, theta) via Black-76 and portfolio **VaR/CVaR** using EWMA volatility shock-and-reprice
- Built a **reversal-probability framework** on NIFTY options book data using session timing, price velocity, and ATR-normalized stretch, trying to isolate sessions with higher reversal rates
- Designed and backtested an intraday de-risking overlay in Python -- evolved a single-trigger partial booking rule into a laddered, velocity-scaled **multi-level system**, improving risk-efficiency on unseen data.

PROJECTS

RISC-V Cache Simulator | Computer Architecture Course

- Implemented a C++ program to support **cache and memory simulation of RISC-V** assembly code
- Added various functionalities like **break points,stacks** for various subroutines,storing register values, **data access** from memory address
- Included hit rate,dirty/clean blocks and customisation of parameters like **replacement policies** (FIFO,LRU,Random) ,**Write Policies** (Write Back,Write Through) , **Associativity** , block size and cache size.

JADE: Distributed/HPC DSL | Compilers Course Project | C++

- Designed JADE, a domain-specific language for distributed computing, with role/task abstractions, communication primitives, static type checking, and deadlock checks.
- Built the lexer-to-C++/MPI compiler pipeline with automatic role-to-rank mapping and demonstrated distributed matrix multiplication/Sudoku workloads.

CCIRP -- Multi-Channel Campaign Platform | Software Engineering Course

- Built a **full-stack campaign management** platform (Next.js/TypeScript frontend + backend API) supporting email,

SMS, and WhatsApp outreach with a 5-step campaign creation workflow
- Implemented **AI-assisted content generation** using an LLM API with **SSE streaming** for real-time chat-based campaign editing
- Developed a **drag-and-drop template editor** with dynamic audience segmentation, an analytics dashboard, and spam-check validation before send

Bank Marketing Prediction | Business Analytics Course | Python

- Built and compared **6 classification models** (Logistic Regression, LDA, Naive Bayes, Decision Tree, Random Forest, Gradient Boosting) on 41K+ customer records to predict term-deposit subscription, selecting **Random Forest** via 5-fold cross-validated ROC-AUC
- Identified and eliminated **data leakage** from a highly predictive feature, demonstrating its inflation effect
- Used **permutation importance, threshold analysis, and lift curves** to convert model output into a lead-prioritization policy where the **top 20% of scored customers captured 65.7% of conversions**

RELEVANT COURSES

- **Mathematics**:- Probability, Linear Algebra, Discrete Maths, Calculus, Intro to Stats, Number Theory and Algebra
- **CS**:- Data Structures, Algorithms, Computer Architecture, Operating systems, Database Management, Theory of Computation, Compilers, Networks
- **ML**:- Foundations of ML, Deep Learning, Reinforcement Learning, Convex Optimisation,

POSITIONS OF RESPONSIBILITY

- **OCS Coordinator**: Assisted in pitching and coordinating with companies for placement and internship recruitment
- **Teaching Assistant** for **4 courses** over the past 2 years: Intro to Programming (ID1063), Probability and Random Variables (AI1110), Theory of Computation (CS2030), and Data Structures and Algorithms (ID2230); assisted with **invigilation, script evaluation,** and **resolving student queries** during classes.
- **Dept Representative** DIESTA 2025: CSE batch representative in Inter-Departmental Fest